Pole Locations 1

EGR 345

GVSU

Pole Locations Intro Questions

- what is a pole location?
- why are pole locations so important in system dynamics and feedback control?
- how do ζ and ω_n relate to second-order pole locations?
- how do ζ and ω_n affect the step response of a system?
- how do I quantify the step response of a system?

What is a pole?

→ value of s where the denominator goes to zero

What are the poles of a second-order, under-damped TF?

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$a = 1$, b, c

quadratic formula

$$s = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$s = \frac{-2\zeta\omega_n \pm 2\omega_n\sqrt{\zeta^2 - 1}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

- $\zeta > 1$ overdamped — $s = p_1, p_2$
- $\zeta = 1$ critically damped
- $\zeta < 1$ underdamped

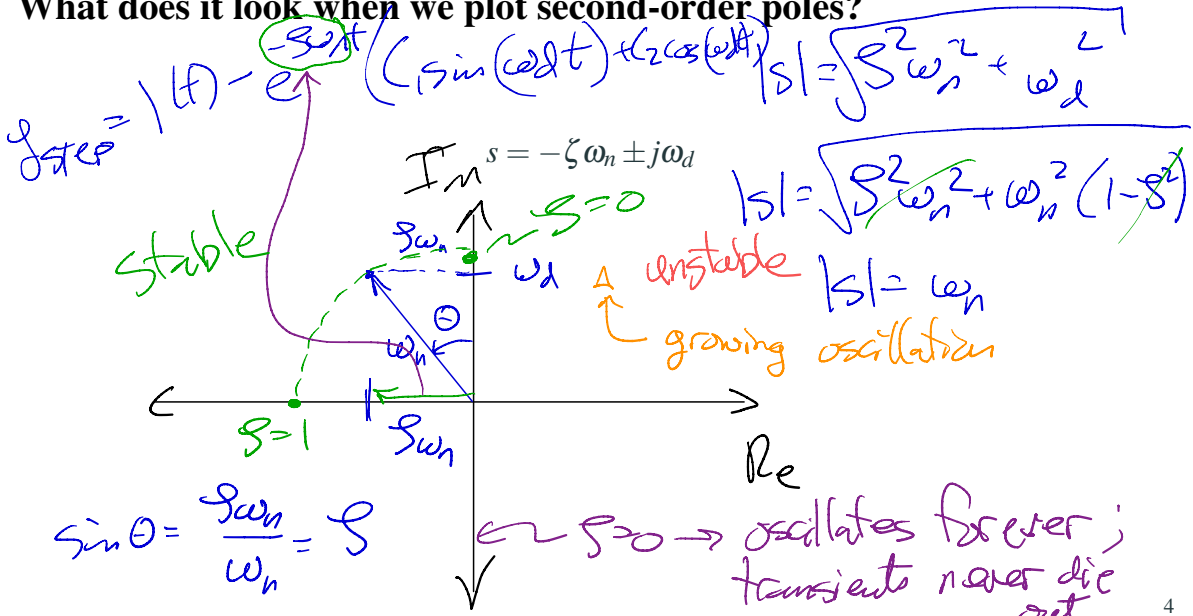
if $\zeta < 1$

$$\sqrt{\zeta^2 - 1} = j\sqrt{1 - \zeta^2}$$

j ← factor out -1

$$s = -\zeta\omega_n \pm j\underbrace{\omega_n\sqrt{1-\zeta^2}}_{\omega_d} = -\zeta\omega_n \pm j\omega_d$$

What does it look when we plot second-order poles?

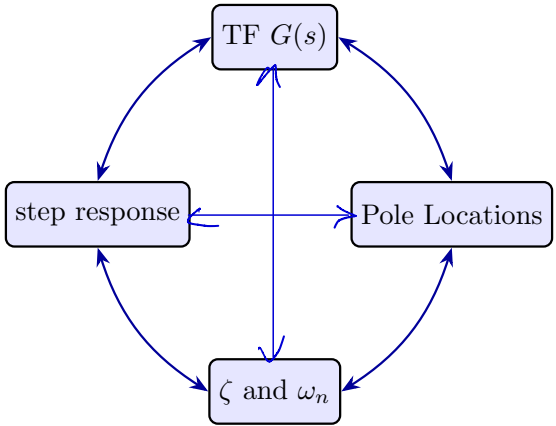


Learning Goals for Pole Locations

- given a sketch of pole locations for second-order TFs:
  - estimate ζ and ω_n
  - roughly sketch the step response in your head or “by hand”
- given ζ and ω_n:
  - plot the pole locations on a real vs. imag. axis
  - sketch the step response
- given the step response of a TF:
  - estimate ζ and ω_n
  - sketch the pole locations

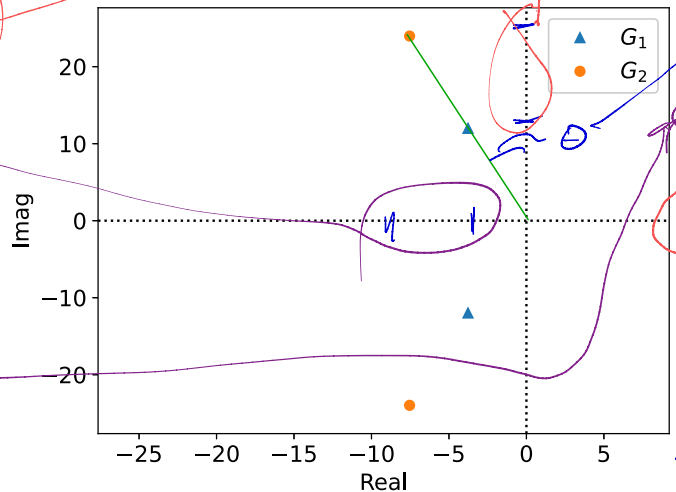
Pole Locations Cycle

- given any *one* of these, we can find *all* the others

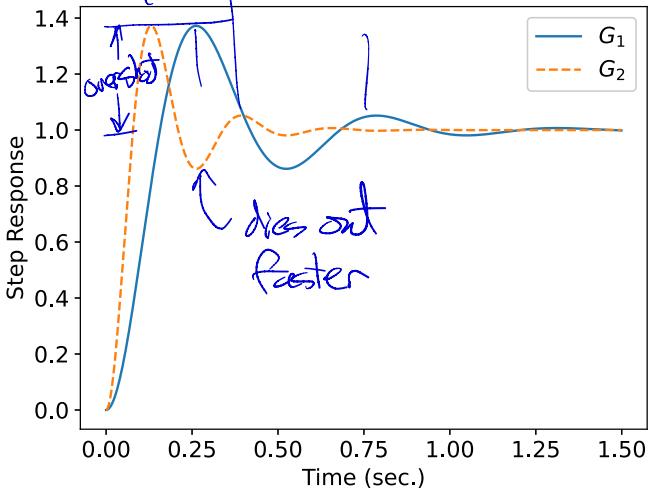


Estimating Step Response from Pole Locations

- how do the step responses compare for G_1 and G_2?

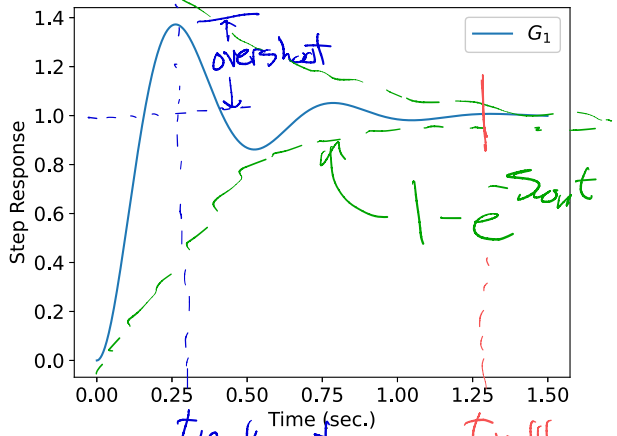


Answer Check for Step Responses of G_1 and G_2



Quantifying a Step Response

- how do we quantify a "typical" second order step response?



Settling time (t_s)

- time from step input until the oscillations essentially

$$e^{-\zeta\omega_n t_s} = 0.02 \text{ or } 0.01$$

$$-\zeta\omega_n t_s = \ln 0.02 \text{ or } \ln 0.01$$

for 2%: $t_s = \frac{3.91}{\zeta\omega_n}$ for 1% $t_s = \frac{4.6}{\zeta\omega_n}$

How do ζ and ω_n connect to our step response measures?

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

assume $0 < \zeta < 1$

- Find $y_{step}(t)$

$C_1 = 1$ (F.V.T. see next slide)

$$Y_{step} = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{C_1}{s} + \frac{C_2\omega_d + C_3(s+a)}{(s+a)^2 + \omega_d^2}$$

$C_3 = -1$ $\quad a = \zeta\omega_n$

Final Value Theorem

$$\lim_{t \to \infty} y(t) = \lim_{s \to 0} s \cdot Y(s)$$

For a step response, $Y(s) = \frac{1}{s} \cdot G(s)$

$$\lim_{t \to \infty} y_{step}(t) = \lim_{s \to 0} s \cdot \frac{1}{s} G(s) = G(0)$$

$$y_{\text{STEP Final Value}} = G(0)$$